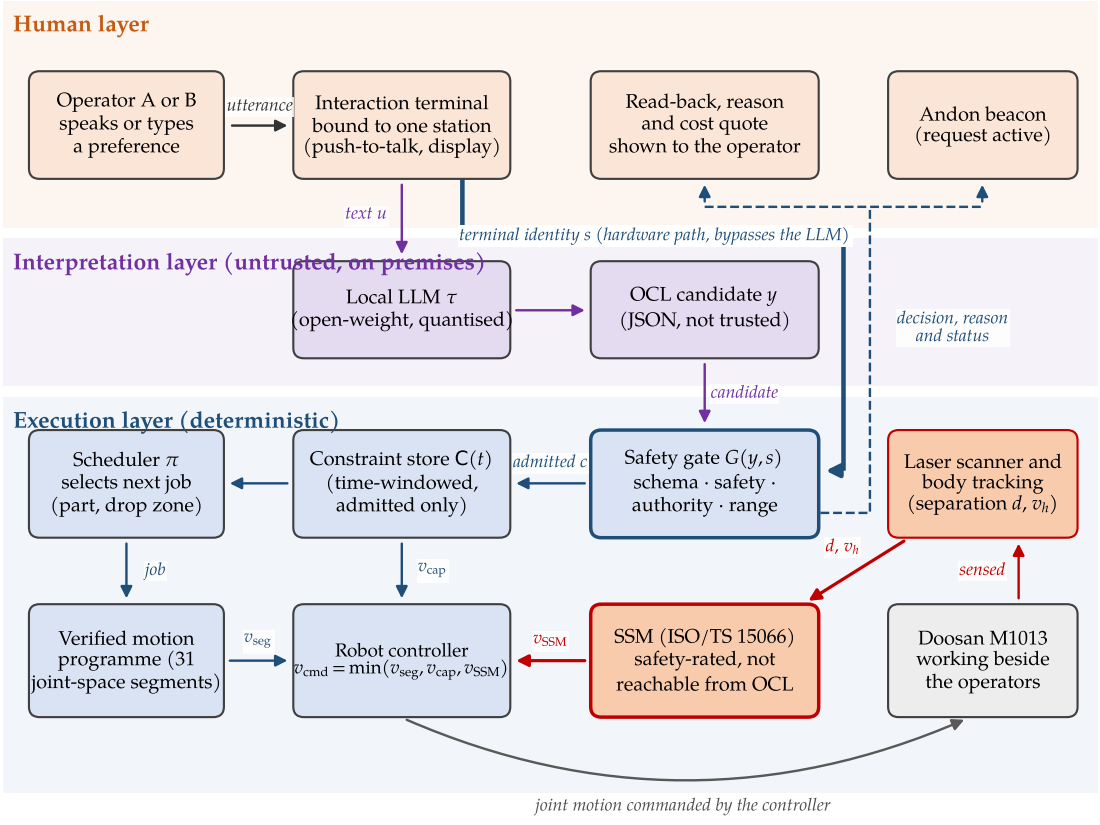
Human layer
Operator A or B speaks or types a preference
utterance
Interaction terminal bound to one station (push-to-talk, display)
Read-back, reason and cost quote shown to the operator
Andon beacon (request active)
text u
terminal identity s (hardware path, bypasses the LLM)
Interpretation layer (untrusted, on premises)
Local LLM τ (open-weight, quantised)
OCL candidate y (JSON, not trusted)
decision, reason and status
candidate
Execution layer (deterministic)
Scheduler π selects next job (part, drop zone)
Constraint store C(t) (time-windowed, admitted only)
admitted c
Safety gate G(y, s) schema · safety · authority · range
Laser scanner and body tracking (separation d, v_h)
d, v_h
sensed
job
v_cap
Verified motion programme (31 joint-space segments)
v_seg
Robot controller v_cmd = min(v_seg, v_cap, v_SSM)
v_SSM
SSM (ISO/TS 15066) safety-rated, not reachable from OCL
Doosan M1013 working beside the operators
joint motion commanded by the controller